Operational Audit & AI Reconciliation Strategy

Author: AI Operations Associate Candidate | Date: 30 June 2026 Snapshot | Focus: Zero-Headcount Optimization

1. DEFINE THE PROBLEM: DATA OBSERVATIONS vs. REPORTED SYMPTOM

Leadership's stated symptom is that 'escalations have been climbing for two months while refund volume is flat.' The underlying diagnosis is an operational disconnection between the two independent trackers rather than external volume growth.

Three distinct empirical observations explain the escalation pattern while refund request volume remains steady (~145/month):
1. Gap Class B (Closed in Support, No Finance Record): Support marked tickets 'Closed' with notes like 'refund processed, informed agent'. However, no corresponding record exists in Finance Tracker. This rate increased from 2.1% in March to 18.7% in May and 33.0% in June. Across the snapshot, exactly 69 closed tickets with no Finance record exist, totaling INR 1,080,700.
2. Finance Processing Turnaround & Documentation: Turnaround measured from Finance received date to payout date on completed records increased from an average of 2.67 days in April to 6.52 days in May (unpaid refunds are excluded from this calculation). Additionally, an approver was recorded on almost no June payouts because refunds above ~INR 20,000 (the inferred approval tier) were mostly still pending; cause not visible in the data.
3. Recorded Deductions on Disbursements: When payouts occurred, Finance recorded Deductions (averaging INR 1,500-2,600). Support records do not document these deductions, leading travel agents to escalate regarding differences between requested and credited amounts.

2. OPERATIONAL ANALYSIS: EMPIRICAL METRICS ACROSS 5 MONTHS

Reconciling 733 unique Support rows and 680 Finance rows reveals 749 unique refund cases. Request volume remained steady across all 5 months (Feb: 144, Mar: 152, Apr: 147, May: 148, Jun: 142). Below is the verified monthly operational summary:

Month	Requests	Support Clsd	Gap B (Count)	Gap B Rate	Avg Proc Time	Done Count	Approver %
Feb 2026	144	137	4	2.9%	2.89 days	111	32 (28.8%)
Mar 2026	152	146	3	2.1%	2.87 days	154	46 (29.9%)
Apr 2026	147	145	6	4.1%	2.67 days	139	41 (29.5%)
May 2026	148	139	26	18.7%	6.52 days	120	29 (24.2%)
Jun 2026	142	91	30	33.0%	4.88 days	59	1 (1.7%)

Footnote: Unpaid refunds are excluded from turnaround calculations. Approver share denominator = all Finance Refund Done rows by received month.

Key Takeaway from Table:
While request volume stayed steady (~145/mo), completed disbursements recorded in Finance for June requests were 59, while Support closed tickets with no Finance record reached 30 (33.0% of Support closures in June). This explains the concentration of agent chases in May and June.

2. OPERATIONAL ANALYSIS (CONT.): TRACKER COMPARISONS & ROOT CAUSES

The two trackers agree on 664 refunds, with 571 completed payouts in Support, but show specific variances in other records:
- Finance Only (Class H - 16 refunds): Finance recorded INR 141,800 in disbursements with no ticket in Support_Tracker.
- Status Variances (Gap C & D - 18 refunds): 12 tickets are Closed in Support but Pending Payout in Finance (INR 186,000 Support sum). 6 tickets are Closed in Support but Declined in Finance (INR 94,800 Support sum).
- Amount Differences (Gap Class E - 47 refunds): Support and Finance record different amounts for 47 completed payouts in Support. Deduction exactly explains all 47 of these differences (Support Refund Amount minus Amount Paid equals recorded Deduction).
- Duplicate Rule: Conflicting duplicate ticket IDs in Support were resolved by keeping the latest Last Updated; on a tie, keeping the row whose Refund Amount equals Finance Amount Paid or Amount Paid + Deduction; otherwise keeping the later row in the sheet.

Root Cause Classification of All 155 Logged Escalations:

Root Cause Category	Count	Share (%)	Data Observation & Action
paid and matched	57	36.8%	Payout recorded in Finance matching request; agent inquired regarding status confirma
Support closed/no Finance	31	20.0%	Gap Class B: Ticket marked Closed on Support, no corresponding Finance record.
never logged (no usable ref)	29	18.7%	Informal inquiries with no usable reference found in operational trackers.
paid but amount differs	18	11.6%	Disbursement matches Amount Paid, but agent expected gross amount without deduct
Finance pending/Support not closed	8	5.2%	Request in Finance processing queue while Support ticket remains active.
declined (unexplained)	6	3.9%	Record declined in Finance; decline details not documented on Support side.
Support closed/Finance pending	6	3.9%	Support ticket marked Closed while Finance record remains in Pending Payout.

Validation Wishlist (What to validate with 2 weeks more access):
1. Finance Approval Workflow: Review workflow logs to verify inferred INR 20k approval threshold and understand pending large refunds.
2. Deduction Schedule Table: Ingest standardized cancellation fee tables to verify deductions prior to payout submission.
3. Informal Request Inboxes: Access incoming email and WhatsApp logs to measure volume of unlogged agent inquiries.

3. DESIGN A SOLUTION: PROCESS & SINGLE SOURCE OF TRUTH (ZERO HEADCOUNT)

Under the no-new-headcount constraint, BharatTrip must link existing workflows into a single source of truth without adding manual checkers:

- Single Unified Table: Support and Finance retain operational interfaces, but read/write to a shared backend. Support status is limited to 'New', 'Working', and 'Verified & Forwarded'. 'Closed' is a system-generated state triggered strictly when Finance records 'Payout Status = Refund Done'.
- Automated Reconciler & Action Queue (core/reconcile.py): Daily reconciliation identifies discrepancies, automatically routing the 69 Gap B records to Finance and ranking open tasks by Priority Score (Amount x Age).
- Proactive Dispatch Notification: On payout execution, automated notification sends the agent the bank transaction reference and an itemized breakdown (Gross Amount, Deduction, Net Disbursed), addressing the pattern behind 57 status queries and 18 deduction disputes.

3. DESIGN A SOLUTION (CONT.): AUTOMATION TRADE-OFFS & AI PROTOTYPE

Component	Operational Choice	Justification & Risk Control
Matching & Reconciliation	100% Deterministic Code (NO LLM)	Zero financial hallucination risk. Ledger math and status joins remain determin
Deduction Verification	Deterministic Rule Engine	Validates whether Refund Amount minus Amount Paid equals Deduction exac
Informal Intake (WhatsApp/Email)	AI / LLM Layer (core/intake.py)	Normalizes unstructured English/Hinglish text into structured JSON schema.
Agent Status Response	Automated Grounded Drafts	Strictly grounded in ledger records. Never invents an estimated turnaround (E
Disbursement Authorization	Deliberately Kept Manual (Finance)	Banking authorization remains with Finance personnel.
Decline Communication	Manual Support Touchpoint	Maintains partner agency communication on contested cancellations.

AI Prototype Implementation:

We built a functional platform backed by modular deterministic code (core/clean.py, core/reconcile.py), FastAPI backend (api.py), and an AI intake engine (core/intake.py):

1. Executive Overview: Visualizes Gap B trends (2.9% to 33.0%) and turnaround on paid refunds.
2. Action Queue & SOT Ledger: Filterable ledger ranking items by priority score with Excel export.
3. Escalation Diagnostics: Categorization across the 8 root cause classes.
4. AI Intake Assistant: Extracts fields from raw messages, queries the ledger, and drafts grounded replies. Enforces unique matching (requests clarification on multiple candidates, routes internal notes to Gap B reports, and assigns non-colliding RF-NEW-### IDs for new inquiries). Built-in zero-key fallback ensures reliable operation without API keys.

5. MEASURE: 30-DAY SUCCESS METRICS & TARGET RESULTS

Metric to Watch	June Baseline	30-Day Target	Operational Significance
1. Gap B Creation Rate	33.0% (30 tickets)	< 2.0% (< 3 tickets)	Direct measure of Support-Finance handoff synchronization.
2. Escalation Rate	35 per 100 refunds	< 8 per 100 refunds	Proactive deduction breakdowns eliminate repeated status queries.
3. Backlog Aging (>14 days)	69 tickets > 30 days	0 tickets > 14 days	Action Queue prioritizes aged high-value items.
4. Intake Auto-Logging Share	~0% (all manual)	> 90% logged	All off-tracker email/WhatsApp requests logged with RF-NEW-###.

Target Result After 1 Month:

- Unrecorded Discrepancy Elimination: Weekly complaints of unrecorded requests drop toward zero.
- Backlog Resolution: The 69 Gap Class B refunds (INR 1,080,700) audited and queued for payment.
- Escalation Volume: Overall escalation rate drops from ~35 per 100 refunds to under 8 per 100 refunds.